

## ORIGINAL ARTICLE

# Cost-Effectiveness of Sodium-Glucose Co-Transporter-2 Inhibitors and Angiotensin Receptor-Nepriylsin Inhibitors by Ejection Fraction and Drug Pricing in the United States: A Systematic Review

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**BACKGROUND:** Sodium-glucose co-transporter 2 (SGLT2) inhibitors and angiotensin receptor-nepriylsin inhibitors (ARNI) are heart failure (HF) therapies selected for price negotiations under the Inflation Reduction Act. This study aimed to summarize the cost-effectiveness of SGLT2 inhibitors and ARNIs for HF, examine how cost-effectiveness varies by ejection fraction (EF) and drug price, and estimate the effect of negotiated prices on cost-effectiveness.

**METHODS:** A systematic literature search identified cost-effectiveness analyses of SGLT2 inhibitors and ARNIs versus standard of care for the treatment of HF from a US perspective, published through 2025. Analyses were stratified by HF with reduced EF (<40%, HFrEF), moderately reduced EF (40% to 49%, HFmrEF), and preserved EF (≥50%, HFpEF). Incremental cost-effectiveness ratios were estimated at negotiated drug prices from base-case, threshold, and sensitivity analyses of drug pricing. Key drivers of cost-effectiveness were identified from the top 3 variables in 1-way sensitivity analyses.

**RESULTS:** Of 821 studies identified, 16 were included: 11 HFrEF, 2 HFrEF/HFmrEF/HFpEF, and 3 HFmrEF/HFpEF. Across all EF categories, incremental cost-effectiveness ratios (2024 US dollars) versus standard of care ranged from \$59 600 to \$187 100/quality-adjusted life year (QALY) gained with SGLT2 inhibitors, \$24 400 to \$92 300 with ARNIs, and \$69 300 to \$91 500 with SGLT2 inhibitors + ARNIs. Incremental cost-effectiveness ratios in studies of HFrEF were <\$120 000/QALY gained and all but 1 were ≥\$120 000/QALY gained in studies of HFmrEF/HFpEF. At negotiated drug prices across all EF categories, the incremental cost-effectiveness ratio was estimated to be \$47 800/QALY gained with SGLT2 inhibitors, \$39 900/QALY gained with ARNIs, and \$44 400/QALY gained with SGLT2 inhibitors+ARNIs. The key drivers of cost-effectiveness included drug price (17 studies), cardiovascular death risk with SGLT2 inhibitors and ARNIs (15 studies), and duration of intervention effectiveness (5 studies).

**CONCLUSIONS:** At negotiated prices resulting from the Inflation Reduction Act, SGLT2 inhibitors and ARNIs seem cost-effective in HFrEF, with projected cost-effectiveness in HFmrEF/HFpEF dependent on price reductions and less certain treatment-effect estimates.

**Key Words:** cost-effectiveness analysis ■ systematic review ■ heart failure

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WHAT IS KNOWN

- Sodium-glucose co-transporter 2 inhibitors and angiotensin receptor-neprilysin inhibitors are costly guideline-directed medical therapies for the treatment of heart failure.
- Sodium-glucose co-transporter 2 inhibitors (dapagliflozin and empagliflozin) and angiotensin receptor-neprilysin inhibitors (sacubitril/valsartan) were among the medications selected for Centers for Medicare & Medicaid Services price negotiations under the Inflation Reduction Act.

WHAT THE STUDY ADDS

- This systematic review summarizes the existing evidence on the cost-effectiveness of sodium-glucose co-transporter 2 inhibitors and angiotensin receptor-neprilysin inhibitors in the treatment of heart failure in the United States across the spectrum of left ventricular ejection fraction.
- Sodium-glucose co-transporter 2 inhibitors and angiotensin receptor-neprilysin inhibitors demonstrate consistent cost-effectiveness in heart failure with reduced ejection fraction, whereas cost-effectiveness in heart failure with mildly reduced or preserved ejection fraction is variable and dependent on drug pricing and underlying treatment-effect assumptions.
- This analysis provides a framework to help decision-makers estimate cost-effectiveness as generic products become available and are more widely used, and to inform future heart failure therapy price negotiations.

Nonstandard Abbreviations and Acronyms

|               |                                                         |
|---------------|---------------------------------------------------------|
| <b>ARNI</b>   | angiotensin receptor-neprilysin inhibitor               |
| <b>CMS</b>    | Centers for Medicare and Medicaid Services              |
| <b>HF</b>     | heart failure                                           |
| <b>HFmrEF</b> | heart failure with moderately reduced ejection fraction |
| <b>HFpEF</b>  | heart failure with preserved ejection fraction          |
| <b>HFrEF</b>  | heart failure with reduced ejection fraction            |
| <b>ICER</b>   | incremental cost-effectiveness ratio                    |
| <b>LVEF</b>   | left ventricular ejection fraction                      |
| <b>QALY</b>   | quality-adjusted life year                              |
| <b>SGLT2</b>  | sodium-glucose co-transporter 2                         |
| <b>SOC</b>    | standard of care                                        |
| <b>USD</b>    | US dollars                                              |

Heart failure (HF), which affects >6.7 million US adults, significantly impacts quality of life and is associated with frequent hospitalizations and increased

mortality.<sup>1</sup> As the US population ages, HF is expected to become a growing health and economic burden in the United States, with costs projected to reach \$70 billion per year by 2030.<sup>2,3</sup> A key driver of costs in HF is related to relatively novel therapies for management.

There are 2 classes of effective yet costly guideline-recommended therapies for HF: sodium-glucose co-transporter 2 (SGLT2) inhibitors and angiotensin receptor-neprilysin inhibitors (ARNI). However, the efficacy of each class varies by left ventricular ejection fraction (LVEF). Trials of both SGLT2 inhibitors (dapagliflozin and empagliflozin) and ARNIs (sacubitril/valsartan) consistently demonstrated a significant reduction in cardiovascular mortality in patients with HF with reduced ejection fraction (HF<sub>r</sub>EF; LVEF <40%).<sup>4–6</sup> Among those with HF with mid-range and preserved ejection fraction (HF<sub>mr</sub>EF, LVEF 40% to 49%; HF with preserved ejection fraction [HF<sub>p</sub>EF], LVEF ≥50%), a reduction in cardiovascular mortality was found only in meta-analyses of SGLT2 inhibitor trials.<sup>7–9</sup> Trials of ARNIs among patients with LVEF ≥45% found reduced cardiovascular mortality only in secondary subgroup analyses.<sup>10</sup>

Economic evaluations of SGLT2 inhibitors and ARNIs for HF have shown that their cost-effectiveness is associated with LVEF and drug price. In an attempt to lower drug prices in the United States, Congress passed the Inflation Reduction Act in 2022, allowing the Centers for Medicare & Medicaid Services (CMS) to negotiate maximum allowable drug prices.<sup>11</sup> In 2024, dapagliflozin, empagliflozin, and sacubitril/valsartan were among the first drugs selected to be subject to price negotiations. The negotiated 30-day supply prices, which took effect in 2026, were \$179 for dapagliflozin, \$197 for empagliflozin, and \$295 for sacubitril/valsartan.<sup>12</sup>

The aim of this review is to summarize the existing evidence on the cost-effectiveness of SGLT2 inhibitors and ARNIs for the treatment of HF, examine how economic value varies by LVEF and drug price, and estimate the effect of CMS-negotiated prices on cost-effectiveness.

METHODS

This systematic review was performed according to a study protocol developed a priori following the recommendations from the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (Tables S1 and S2).<sup>13</sup> The protocol is available upon request. Deviations from the protocol are reported in the Supplemental Material. The data that support the findings of this study are available within the article and Supplemental Material, including all data extracted from each study (Table S3). Additional information regarding the data and analyses are available upon reasonable request from the corresponding author. The data used in this study are derived from published literature, and the analysis is not considered human subject research; institutional review board approval was not required.

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## Search Strategy and Selection Criteria

A literature search was performed in PubMed and Embase (OVID) on February 28, 2026, using a prespecified search strategy (Supplemental Methods). The search strategy was designed and developed with librarian consultation and included medical subject headings, keywords, drug names, and relevant clinical trial names included in the title or abstract of articles. Search terms included cost-effectiveness, cost benefit, cost utility, incremental cost-effectiveness ratio (ICER), SGLT2 inhibitor, ARNI, and relevant drug names. Synonyms and supplementary search terms were identified using the medical subject headings tree structure and included in the final search. The search was restricted to publications through 2025. In PubMed, articles were restricted to those published in English. In Embase (OVID), filters were used to exclude abstracts, reviews, editorials, commentaries, and retracted articles. All identified studies were uploaded into Covidence systematic review software (Veritas Health Innovation, Melbourne, Australia) for review.<sup>14</sup>

Cost-effectiveness analyses, cost-benefit analyses, cost-utility analyses or any model-based analysis that evaluated the cost-effectiveness of treatment of HF with regimens including SGLT2 inhibitors and ARNIs were included. As an objective was to examine the effect of CMS-pricing on cost-effectiveness, we excluded studies on medications without negotiated CMS prices. To improve comparability, studies that did not include a summative outcome in the form of an ICER were excluded. To examine the drug prices in the United States, studies from a non-US health care perspective or focused on the treatment of non-HF conditions such as type 2 diabetes and chronic kidney disease were excluded. Studies that were systematic or narrative reviews, editorials, or commentaries were also excluded.

## Study Selection Process and Review for Inclusion

The titles and abstracts of all studies identified from the literature search were uploaded into Covidence for review. All titles and abstracts were screened using the proposed inclusion and exclusion criteria by 2 blinded reviewers (T.A. and B.B.), and conflicts were resolved by consensus with a third reviewer (L.C.). Full-text review was completed independently by 2 reviewers (T.A., B.B., and L.C.), and discrepancies were resolved by agreement with a third reviewer. For studies included in full-text review, prioritized reasons for exclusion were specified: (1) wrong study characteristics (reviews, editorials, commentaries); (2) non-US health care perspective; or (3) wrong study design, wrong intervention, wrong comparators, wrong outcomes, or wrong setting. Studies excluded during this step were categorized under the highest priority exclusion criteria met.

## Data Collection

Two reviewers independently extracted data from each included study manually into Microsoft Excel (Microsoft Corporation, Redmond, WA; T.A., B.B., and L.C.). Data extracted included study characteristics, ejection fraction range studied, costs, drug cost inputs, quality-adjusted life years (QALYs), ICERs, willingness-to-pay thresholds, probabilistic sensitivity analyses, 1-way sensitivity analyses, and funding source (Table 1; Table S3).<sup>15–30</sup> Data were then reviewed for discrepancies (T.A.),

which were resolved by agreement among 3 reviewers (T.A., B.B., and L.C.) and referenced to the original published texts.

## Data Analysis

The number of cost-effectiveness analyses published for each drug class and LVEF studied was evaluated. Descriptive comparisons of the studies were performed with regard to model structure, outcomes, subgroup analyses, and probabilistic and 1-way sensitivity analyses.

## Cost-Effectiveness Outcomes

Cost-effectiveness outcomes, including total and incremental health care costs, drug costs, QALYs, and ICERs (cost per QALY gained), were identified and summarized. Analyses were stratified by drug class (SGLT2 inhibitor, ARNI, or both) and HF LVEF subtype (HFrEF, HFmrEF, and HFpEF). Total health care costs, drug costs, and ICERs were inflated to 2024 US dollars (USD) using the health care component of the US Bureau of Economic Analysis Personal Consumption Expenditure Price Index.<sup>31</sup> A cost-effectiveness threshold of <\$120 000 per QALY was used to categorize ICERs as cost-effective compared with standard of care (SOC), based on the recommendations of the 2025 American Heart Association/American College of Cardiology Statement on Cost/Value Methodology in Clinical Practice Guidelines.<sup>32</sup> Linear mixed-effects models were used to summarize how the ICER for SGLT2 inhibitors, ARNIs, and SGLT2 inhibitors + ARNIs versus SOC varied with drug price. For each study, the base-case estimate and, when available, upper and lower bound estimates (derived from 1-way sensitivity analyses or threshold analyses) were used. If both threshold and sensitivity analyses were reported, the least extreme values were used to determine the upper and lower bounds included in analyses. All models included a fixed effect for drug price to estimate the mean sensitivity response across studies and a random intercept for study to account for between-study differences in base-case estimates. Random slopes were not included because not all studies reported upper and lower bound estimates associated with drug cost changes. The fitted fixed-effect lines were used to represent the average sensitivity response across studies. Stratified analyses were performed to examine differences within LVEF. When data allowed, the fitted fixed-effect models were used to generate estimated ICERs versus SOC at CMS-negotiated drug prices. The fitted fixed-effects models were also used to generate 95% CIs for the predicted ICER values across the range of drug costs. The 95% CIs were derived from 500 parametric bootstrap samples of the linear mixed-effects model. For each bootstrap replicate, fixed-effects predictions were generated, and percentile-based intervals were calculated from the 2.5th and 97.5th percentiles of the bootstrapped predictions. In the secondary analysis, the results of any subgroup analyses by LVEF, including the ICER and any upper or lower bound estimates associated with drug cost changes, were also included when fitting the models.

## Drivers of Cost-Effectiveness

To identify the key drivers of cost-effectiveness of SGLT2 inhibitors and ARNIs, the 3 variables to which each model was reported to be the most sensitive in 1-way sensitivity analyses were extracted. Variables were categorized into cost of the intervention (ie, SGLT2 inhibitor or ARNI) and SOC, cardiovascular death risk (eg, probability or relative risk), cardiovascular

Table 1. Summary of Characteristics and Results of Identified Studies

| Study                          | Year | LVEF (HF subtype)                | Drug: intervention, comparator           | Annual drug cost (2024 USD) | Life years (Undisc.) | QALYs | ICER (2024 USD)  | CHEQUE score*: methods, reporting | Industry funding†              |
|--------------------------------|------|----------------------------------|------------------------------------------|-----------------------------|----------------------|-------|------------------|-----------------------------------|--------------------------------|
| King et al <sup>15</sup>       | 2016 | ≤40% (HFrEF)                     | Sacubitril/valsartan                     | \$5504                      | 9.48                 | 6.59  | \$61 505         | 97.0, 96.0                        | Industry relationship reported |
|                                |      |                                  | Enalapril                                | \$338                       | 8.40                 | 5.83  |                  |                                   |                                |
| Sandhu et al <sup>16</sup>     | 2016 | ≤40% (HFrEF)                     | Sacubitril/valsartan                     | \$5507                      | 7.67                 | 6.33  | \$56 791         | 96.0, 95.0                        | Industry relationship reported |
|                                |      |                                  | Lisinopril                               | \$27                        | 6.98                 | 5.71  |                  |                                   |                                |
| Gaziano et al <sup>17</sup>    | 2016 | ≤40% (HFrEF)                     | Sacubitril/valsartan                     | \$5462                      | 11.08                | 6.80  | \$54 641         | 90.0, 94.5                        | Industry relationship reported |
|                                |      |                                  | Enalapril                                | \$117                       | 9.65                 | 6.02  |                  |                                   |                                |
| Zueger et al <sup>18</sup>     | 2018 | ≤40% (HFrEF)                     | Sacubitril/valsartan                     | \$5868                      | NR                   | 5.05  | \$79 957         | 94.0, 88.5                        | None                           |
|                                |      |                                  | Enalapril                                | \$840                       | NR                   | 4.56  |                  |                                   |                                |
| Gaziano et al <sup>19</sup>    | 2020 | ≤40% (HFrEF)                     | Sacubitril/valsartan                     | \$6383                      | 8.44                 | 6.72  | \$24 419         | 96.0, 96.0                        | Industry funded                |
|                                |      |                                  | Enalapril                                | \$198                       | 6.93                 | 5.49  |                  |                                   |                                |
| Parizo et al <sup>20</sup>     | 2021 | ≤40% (HFrEF)                     | Dapagliflozin                            | \$6287                      | 9.20                 | 5.70  | \$92 523         | 96.0, 96.0                        | Industry relationship reported |
|                                |      |                                  | SOC (ACE inhibitor/ARB/ARNI, BB)         | \$0                         | 8.50                 | 5.20  |                  |                                   |                                |
| Isaza et al <sup>21</sup>      | 2021 | ≤40% (HFrEF)                     | Dapagliflozin                            | \$4637                      | 7.73                 | 5.36  | \$75 545         | 99.0, 97.5                        | Industry relationship reported |
|                                |      |                                  | SOC (ACE inhibitor/ARB/ARNI, BB, MRA)    | \$0                         | 6.82                 | 4.73  |                  |                                   |                                |
| Zheng et al <sup>22</sup>      | 2022 | >40% (HFmrEF, HFpEF)             | Empagliflozin                            | \$4214                      | 9.00                 | 5.00  | \$187 089        | 99.0, 96.0                        | Industry relationship reported |
|                                |      |                                  | SOC                                      | \$0                         | 8.80                 | 4.90  |                  |                                   |                                |
| Nguyen et al <sup>23</sup>     | 2023 | ≤40% (HFrEF)                     | Dapagliflozin                            | \$4988                      | NR                   | 4.88  | \$59 603         | 85.0, 90.0                        | None                           |
|                                |      |                                  | SOC (ACE inhibitor/ARB/ARNI, BB, MRA)    | \$1512                      | NR                   | 4.50  |                  |                                   |                                |
|                                |      | ≤40% (HFrEF)                     | Empagliflozin                            | \$4333                      | NR                   | 4.68  | \$93 692         |                                   |                                |
|                                |      |                                  | SOC (ACE inhibitor/ARB/ARNI, BB, MRA)    | \$1512                      | NR                   | 4.50  |                  |                                   |                                |
| Bhatt et al <sup>24</sup>      | 2023 | ≤60% (HFrEF, HFmrEF, HFpEF)      | Sacubitril/valsartan                     | \$7622                      | 9.72                 | 7.67  | \$82 608         | 97.0, 92.5                        | Industry funded                |
|                                |      |                                  | RAS inhibitor                            | \$198                       | 9.12                 | 7.14  |                  |                                   |                                |
| Cohen et al <sup>25</sup>      | 2023 | >40% (HFmrEF, HFpEF)             | SGLT2 inhibitors                         | \$4730                      | 7.95                 | 5.46  | \$148 215        | 99.0, 99.5                        | Industry relationship reported |
|                                |      |                                  | SOC                                      | \$0                         | 7.73                 | 5.27  |                  |                                   |                                |
| Yan et al <sup>26</sup>        | 2023 | ≤40% (HFrEF)                     | SGLT2 inhibitor + SOC (enalapril/MRA/BB) | \$5471                      | 10.12                | 6.57  | \$76 627         | 99.0, 96.5                        | None                           |
|                                |      |                                  | SOC (ACE inhibitor, BB, MRA)             | \$44                        | 9.25                 | 6.03  |                  |                                   |                                |
|                                |      | ≤40% (HFrEF)                     | ARNI + (MRA/BB)                          | \$7123                      | 10.35                | 6.69  | weakly dominated |                                   |                                |
|                                |      |                                  | SOC (ACE inhibitor, BB, MRA)             | \$44                        | 9.25                 | 6.03  |                  |                                   |                                |
|                                |      | ≤40% (HFrEF)                     | Quad therapy                             | \$12 594                    | 11.26                | 7.25  | \$91 532         |                                   |                                |
|                                |      |                                  | SOC (ACE inhibitor, BB, MRA)             | \$44                        | 9.25                 | 6.03  |                  |                                   |                                |
|                                |      | ≤40% (HFrEF)                     | Quad therapy                             | \$9569                      | 11.26                | 7.25  | \$69 279         |                                   |                                |
|                                |      |                                  | SOC (ACE inhibitor, BB, MRA)             | \$28                        | 9.25                 | 6.03  |                  |                                   |                                |
| Dixit et al <sup>27</sup>      | 2023 | ≤40% (HFrEF)                     | Quad therapy                             | \$10 359                    | 10.99                | 7.30  | \$85 024         | 95.0, 98.5                        | Industry relationship reported |
|                                |      |                                  | ACE inhibitor, BB, MRA                   | \$56                        | 9.26                 | 6.18  |                  |                                   |                                |
| Reifsnider et al <sup>28</sup> | 2024 | ≤40% (HFrEF)                     | Empagliflozin                            | \$6733                      | 5.96                 | 4.36  | \$94 295         | 99.0, 95.0                        | Industry funded                |
|                                |      |                                  | SOC                                      | \$2786                      | 5.77                 | 4.17  |                  |                                   |                                |
| Bhatt et al <sup>29</sup>      | 2024 | ≤40, >40% (HFrEF, HFmrEF, HFpEF) | Dapagliflozin                            | \$6486                      | 7.98                 | 6.57  | \$89 804         | 99.5, 97.5                        | Industry funded                |
|                                |      |                                  | SOC                                      | \$0                         | 7.47                 | 6.04  |                  |                                   |                                |

(Continued)

Table 1. Continued

| Study                     | Year | LVEF (HF subtype) | Drug: intervention, comparator | Annual drug cost (2024 USD) | Life years (Undisc.) | QALYs | ICER (2024 USD) | CHEQUE score*: methods, reporting | Industry funding†              |
|---------------------------|------|-------------------|--------------------------------|-----------------------------|----------------------|-------|-----------------|-----------------------------------|--------------------------------|
| Dixit et al <sup>90</sup> | 2024 | 45–52 (HFmrEF)    | SGLT2 inhibitor + MRA          | \$5089                      | 9.27                 | 6.32  | \$115 900       | 98.5, 99.0                        | Industry relationship reported |
|                           |      |                   | MRA                            | \$18                        | 8.73                 | 5.96  |                 |                                   |                                |
|                           |      | >52 (HFpEF)       | SGLT2 inhibitor + MRA          | \$5089                      | 12.30                | 8.02  | \$144 619       |                                   |                                |
|                           |      |                   | MRA                            | \$18                        | 11.75                | 7.67  |                 |                                   |                                |

The table shows the summarized characteristics and results from the studies identified and included in analysis. All costs and ICERs were inflated to USD 2024. ACE indicates angiotensin-converting enzyme; ARB, aldosterone receptor blocker; ARNI, angiotensin receptor-neprilysin inhibitor; BB,  $\beta$ -blocker; CHEQUE, Criteria for Health Economic Quality Evaluation; EF, ejection fraction; HF, heart failure; HFmrEF, heart failure with mid-range ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; ICER, incremental cost-effectiveness ratio; MRA, mineralocorticoid receptor antagonist; QALY, quality-adjusted life year; SGLT2, sodium-glucose co-transporter 2; SOC, standard of care; Undisc., undiscounted; and USD, US dollars.

\*The CHEQUE tool is a quality assessment tool for cost-effectiveness analysis. It is a 48-item assessment, separated into 24 items describing quality of methods and 24 items reporting quality of reporting. The items are weighted so that each section has a total possible score of 100, depending on the scoring approach used. The degree to which the study meets each item is graded using yes (multiply item weight by 1), no (multiply item weight by 0), or somewhat (multiply item weight by 0.5). The CHEQUE tool is designed to provide a relative ranking of cost-effectiveness analysis studies with a higher score representing a higher quality. The CHEQUE tool does not define a threshold differentiating high- from low-quality cost-effectiveness analysis.

†Industry relationship reported indicates studies in which authors reported relevant industry partnerships, however no direct funding was reported for the study.

hospitalization risk (ie, HF or cardiovascular hospitalizations), duration of effectiveness (ie, how long the intervention was assumed to reduce risk), utility, cost of hospitalizations, and all-cause death risk. All categorized variables were then quantified by the total number of times each category was reported and the number of studies reporting each category as the most sensitive variable.

Methodologic and Reporting Quality Assessment

Methodologic and reporting quality were assessed using the Criteria for Health Economic Quality Evaluation tool.<sup>33</sup> The Criteria for Health Economic Quality Evaluation tool is a 48-item assessment, separated into 24 items describing methodologic quality and 24 items describing reporting quality. The items are weighted so that each section has a total possible score of 100, depending on the scoring approach used. The degree to which the study met each item was graded using yes (multiply item weight by 1), no (multiply item weight by 0), or somewhat (multiply item weight by 0.5). All included studies were independently scored by 2 reviewers (T.A., J.W., B.B., and L.C.), with a third reviewer used in cases of conflict of interest or authorship or to resolve discrepancies.

RESULTS

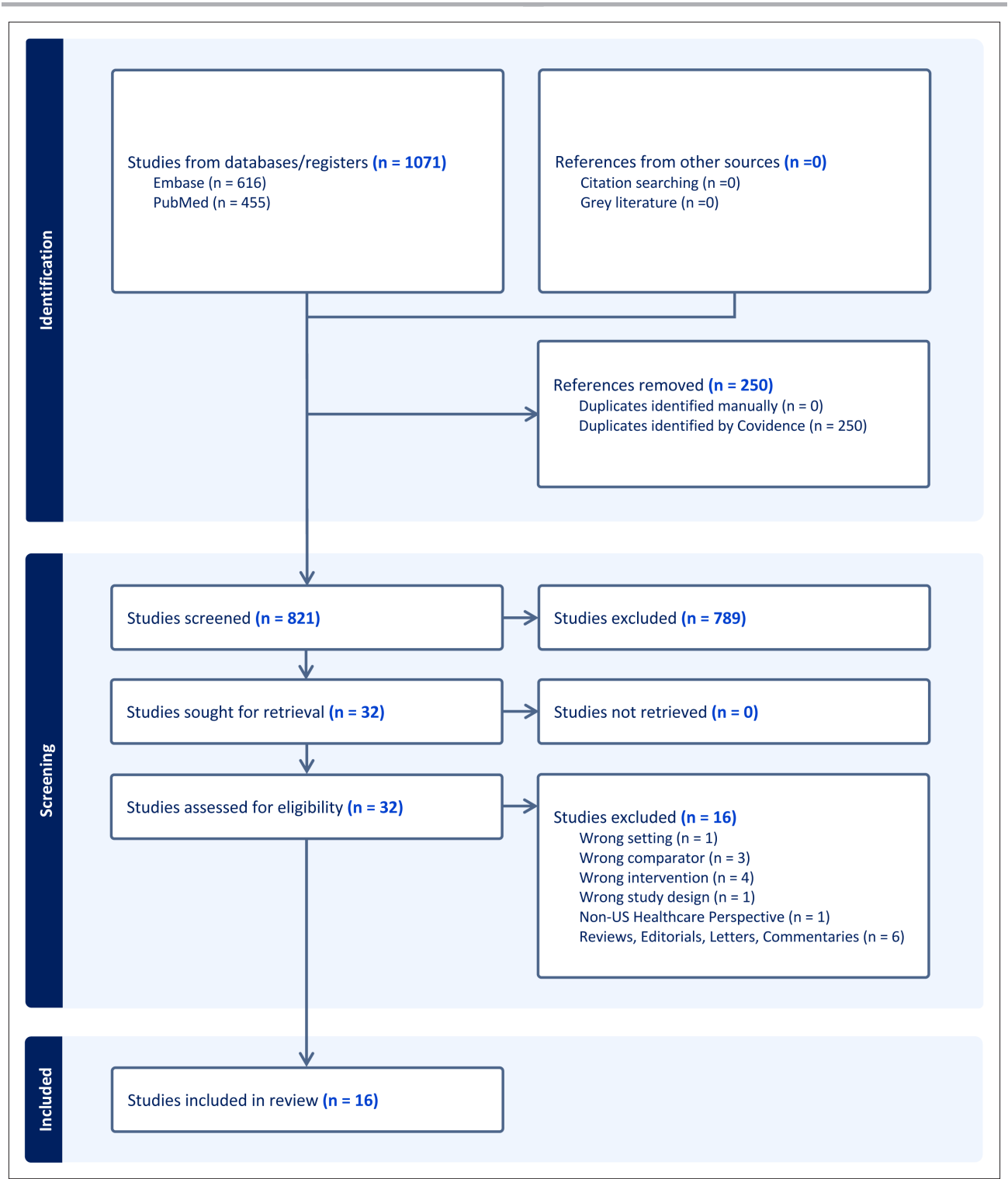
In total, 821 articles were identified from PubMed and Embase (OVID). After initial title and abstract screening, 32 articles underwent full-text review (Figure 1). After excluding full-text articles that did not meet the eligibility criteria, 18 studies were included for data extraction. After full data extraction, 2 articles were excluded from analysis because they compared cost-effectiveness between 2 different SGLT2 inhibitors.<sup>34,35</sup> As an objective was to examine the effect of CMS-negotiated pricing on cost-effectiveness, 2 studies examining sotagliflozin, which was not included in the CMS-negotiated prices, versus SOC were excluded.<sup>36,37</sup> A total of 16 articles were included in the final analysis.<sup>15–30</sup> The final extracted data are included in the Supplemental Materials.

Overall Summary

The details of all included studies can be found in Table 1 and Table S3. Seven studies examined the cost-effectiveness of SGLT2 inhibitors,<sup>20–23,25,28,29</sup> 6 studies examined the cost-effectiveness of ARNIs,<sup>15–19,24</sup> and 3 studies examined both SGLT2 inhibitors and ARNIs (alone and combined).<sup>26,27,30</sup> Most studies analyzed HFrEF populations.<sup>15–21,23,26–28</sup> Two studies analyzed HFpEF as well as HFmrEF and HFpEF populations<sup>24,29</sup> and 3 studies strictly analyzed HFmrEF and HFpEF.<sup>22,25,30</sup> Studies varied in how they defined HFmrEF and HFpEF LVEF ranges.

All studies used a health state transition (Markov) decision model with varying numbers of health states, between 2 and 5. The majority of studies evaluated a 30-year or lifetime time horizon as their primary analysis (Table S3). One study used a 5-year time horizon for the primary analysis and a 30-year time horizon for the secondary analysis—results from the 30-year time horizon were examined for consistency across studies.<sup>18</sup> Studies adopted a societal,<sup>16</sup> health care system,<sup>19,20,22–26,29,30</sup> or payer perspective<sup>15,17,18,20,27,28</sup> (Table S3). Studies sourced drug costs from sources including Micromedex RedBook wholesale acquisition costs,<sup>15–19,24,26,28,29</sup> SSR Health wholesale acquisition costs,<sup>21,25</sup> Department of Veterans Affairs Federal Supply Schedule,<sup>22,23,25–27,29,30</sup> or Medicare Part D drug payments<sup>20</sup> (Table S3). Four studies were industry funded,<sup>19,24,28,29</sup> 3 were not,<sup>18,23,26</sup> and in the remaining studies<sup>15–17,20–22,25,27,30</sup> authors reported some relationship with industry (eg, research funding unrelated to the cost-effectiveness analysis).

All included studies reported the estimated mean treatment costs, mean health care costs, QALYs, and ICERs, and most reported undiscounted life years gained. All studies reported probabilistic and 1-way sensitivity or threshold analyses for their main analyses;



**Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart.**  
The figure shows the PRISMA flowchart showing the number of studies identified, reviewed, and included/excluded at each step of the review process.

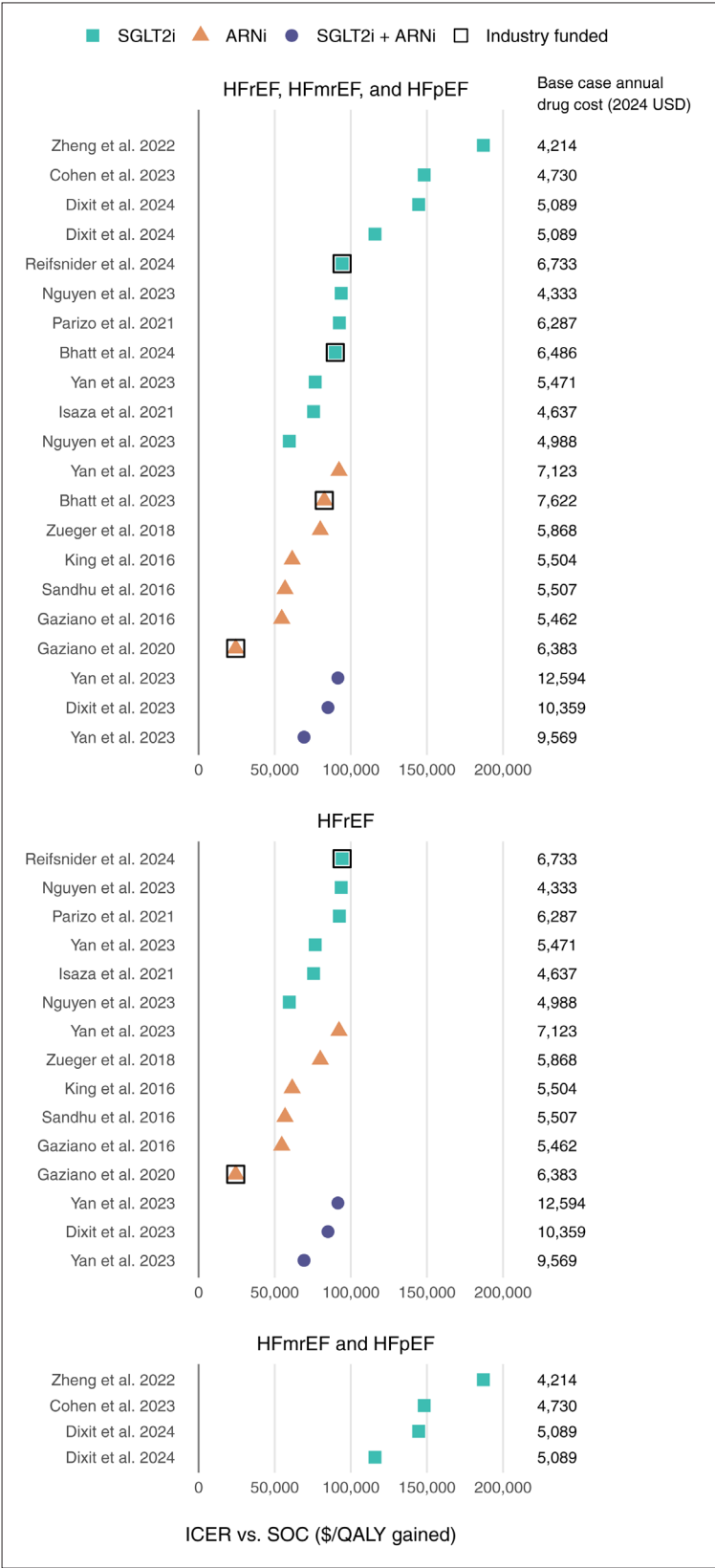
however, not all studies reported upper and lower bounds for the drugs and subgroups included in the summary estimates. There were 2 studies of SGLT2 inhibitors used in HFmrEF and HFpEF that reported results both

with and without SGLT2 inhibitor-associated reductions in mortality to address inconsistencies in significant mortality benefit between the primary trial results and meta-analyses.<sup>22,25</sup>

Cost-Effectiveness Outcomes

Across all LVEF, ICERs versus SOC (2024 USD) ranged from \$59 600 to \$187 100 per QALY gained with

SGLT2 inhibitors, \$24 400 to \$92 300 with ARNIs, and \$69 300 to \$91 500 with SGLT2 inhibitors + ARNIs (Figure 2). In the primary analysis, ICERs versus SOC in studies of HFrEF were all <\$120 000 per QALY



**Figure 2. Incremental cost-effectiveness ratio vs standard of care by ejection fraction.**

The figure shows the incremental cost-effectiveness ratio (ICER) vs standard of care (SOC) stratified by ejection fraction (EF) for studies included in the primary analysis. The points are from the base-case analyses of the included studies. Points encompassed in a square are those funded by industry. The right side of the figure reports the drug cost in 2024 US dollars (USD) associated with the ICER in the base-case analyses. Zheng 2022,<sup>22</sup> Cohen 2023,<sup>25</sup> Dixit 2024,<sup>30</sup> Reifsnider 2024,<sup>28</sup> Nguyen 2023,<sup>23</sup> Parizo 2021,<sup>20</sup> Bhatt 2024,<sup>29</sup> Yan 2023,<sup>26</sup> Isaza 2021,<sup>21</sup> Bhatt 2023,<sup>24</sup> Zueger 2018,<sup>18</sup> King 2016,<sup>15</sup> Sandhu 2016,<sup>16</sup> Gaziano 2016,<sup>17</sup> Gaziano 2020,<sup>19</sup> Dixit 2023.<sup>27</sup> ARNI indicates angiotensin receptor-neprilysin inhibitor; HF, heart failure; HFmrEF, heart failure with mid-range ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; QALY, quality-adjusted life year; and SGLT2i, sodium-glucose co-transporter 2 inhibitor.

gained, and all but 1 were  $\geq \$120\,000$  per QALY gained in studies of HFmrEF and HFpEF. At CMS-negotiated drug prices across all LVEF, the ICER versus SOC was estimated to be \$47 800 (95% CI, \$17 200–\$79 800) per QALY gained with SGLT2 inhibitors (annual cost of dapagliflozin, \$2172), \$39 900 (95% CI, \$21 800–\$60 300) per QALY gained with ARNIs (annual cost of sacubitril/valsartan, \$3589), and \$44 400 (95% CI, \$35 000–\$54 300) per QALY gained with SGLT2 inhibitor + ARNI (Figure 3; Table 2). In HFrEF, we estimated that SGLT2 inhibitors would be cost-effective versus SOC (ICER  $< \$120\,000$  per QALY gained) when priced  $\leq \$8049$  per year, ARNIs when priced  $\leq \$11\,670$  per year, and SGLT2 inhibitor + ARNI when priced  $\leq \$15\,510$  per year (combined). In HFmrEF and HFpEF, only SGLT2 inhibitors had enough data from the primary analysis for estimates; SGLT2 inhibitors would be cost-effective when priced  $\leq \$3631$  per year. The results were quantitatively similar when including reported subgroup analyses by LVEF and excluding studies funded by industry (Figures S1 and S2).

### Drivers of Cost-Effectiveness

All studies performed and reported 1-way sensitivity analyses, with most reporting the top 2 or 3 most sensitive variables. However, several studies lacked comprehensive 1-way sensitivity analyses on all model variables, limiting their analyses to specific input variables. The inputs reported were evaluated to determine the greatest drivers of cost-effectiveness (Figure 4). Drug price and cardiovascular death risk with SGLT2 inhibitors and ARNIs were the variables to which the cost-effectiveness was most sensitive.

### Methodologic and Reporting Quality Assessment

Overall, all studies included scored  $\geq 85$  out of 100 on both the Criteria for Health Economic Quality Evaluation tool methodologic and reporting assessments (Table 1; Table S4). There did not seem to be an association with the ICER and Criteria for Health Economic Quality Evaluation scores regardless of industry funding (Figure S3). The most commonly missed methodologic item was the inclusion of relevant equity or distributional considerations. The most commonly missed reporting item was discussion of any significant ethical implications of the cost-effectiveness results. Other commonly missed reporting items included comprehensive identification of potential consequences (eg, using an impact inventory table), presenting results in a disaggregated format for transparency, and reporting quantities of resources separately from the prices (unit costs) of those resources.

## DISCUSSION

### Summary of Findings

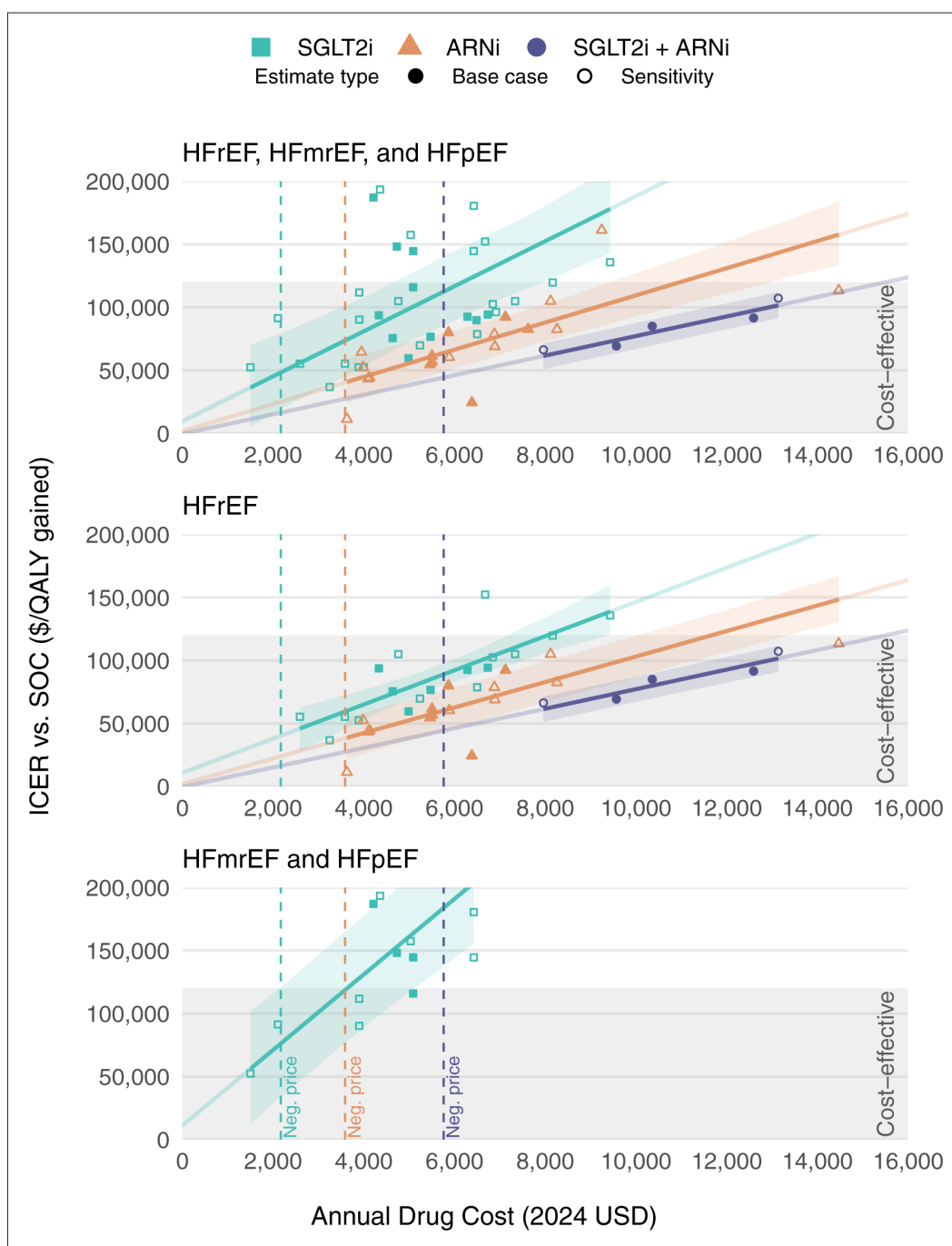
This systematic review summarized the existing evidence on the cost-effectiveness of SGLT2 inhibitors and ARNIs for the treatment of HF in the United States, considering both LVEF and drug prices. Consistent with prior studies, ICERs varied across LVEF ranges and by drug price, with SGLT2 inhibitors and ARNIs more cost-effective at lower LVEF ranges. At Inflation Reduction Act CMS-negotiated annual prices for SGLT2 inhibitors (dapagliflozin, \$2172; empagliflozin, \$2397) and ARNIs (sacubitril/valsartan, \$3589), both SGLT2 inhibitors and ARNIs were estimated to be cost-effective at a willingness-to-pay threshold of \$120 000 per QALY, regardless of LVEF.

### Comparison With Existing Literature

Consistent with this study, prior systematic reviews have determined SGLT2 inhibitors and ARNIs to be cost-effective in the treatment of HFrEF, though none have been specific to a US health care perspective.<sup>38–40</sup> A prior systematic review found SGLT2 inhibitors to be cost-effective in the treatment of HFpEF in some countries, but not in the United States at current drug prices.<sup>38</sup> Compared with the  $< \$120\,000$  per QALY cost-effectiveness threshold used in this study, the 2022 American Heart Association and American College of Cardiology Guidelines for the Management of Heart Failure included value statements on medications for which there were deemed to be high-quality economic analyses.<sup>41</sup> For the treatment of HFrEF, the guidelines classified SGLT2 inhibitors as intermediate economic value (ICER  $\geq \$60\,000$  to  $< \$180\,000$  per QALY gained) and ARNIs as high economic value (ICER  $< \$60\,000$  per QALY gained). For HFmrEF or HFpEF populations, they did not make a statement of economic value for SGLT2 inhibitors.<sup>41</sup> Using the same value framework, our analyses estimate that under current drug pricing (2024 USD), SGLT2 inhibitors would be low (ICER  $\geq \$180\,000$  per QALY gained) to intermediate economic value for the treatment of HFmrEF and HFpEF. However, at the CMS-negotiated prices, which went into effect in 2026, SGLT2 inhibitors would be high value in HFrEF and intermediate value in HFmrEF and HFpEF; and ARNIs would be high value in HFrEF, according to the HF value framework.

### Implications for Future Studies and Health Policy

Cost-effectiveness analyses are not designed to drive individual clinical decisions but rather to assess the economic efficiency of a treatment relative to its clinical benefit, thereby informing health policy. This analysis



**Figure 3. Association of incremental cost-effectiveness ratio vs standard of care with drug price by ejection fraction (EF).**

The figure shows the association between the incremental cost-effectiveness ratio (ICER) vs standard of care (SOC) and annual drug price stratified by EF. The solid points are from the base-case analyses, and the unfilled points are from sensitivity or threshold analyses. The solid lines are the fitted fixed-effect line representing the average sensitivity response across studies, which were estimated using linear mixed-effects models with a random intercept for study. The lighter parts of the lines are the extrapolations beyond the drug prices reported in the studies. The shaded colored bands represent 95% CIs around the estimated fixed-effects regression line and reflect uncertainty in the estimated association between ICER and drug costs across studies. Confidence bands were derived from 500 parametric bootstrap samples of the linear mixed-effects model. For each bootstrap replicate, fixed-effects predictions were generated (excluding random effects), and percentile-based intervals were calculated from the 2.5th and 97.5th percentiles of the bootstrapped prediction. The dotted vertical lines represent the annual drug prices negotiated by the Centers for Medicare and Medicaid Services for dapagliflozin, sacubitril/valsartan, and dapagliflozin + sacubitril/valsartan. The grey shaded region represents when the ICER vs SOC is below the cost-effectiveness threshold of \$120 000 per quality-adjusted life year (QALY) gained. ARNi indicates angiotensin receptor-neprilysin inhibitor; HF, heart failure; HFmrEF, heart failure with mid-range ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; Neg. price, price negotiated by Centers for Medicare and Medicaid Services; Sens./thresh., sensitivity or threshold analysis; SGLT2i, sodium-glucose co-transporter 2 inhibitor; and USD, US dollars.

**Table 2. Estimated Incremental Cost-Effectiveness Ratios of SGLT2 Inhibitors, ARNIs, and SGLT2 Inhibitor + ARNI Versus Standard of Care at Centers for Medicare and Medicaid Services Negotiated Drug Prices by Drug and Ejection Fraction**

| Drug                            | CMS price* | Estimated ICER (95% CI)†      | Estimated drug price for \$120 000 per QALY gained‡ |
|---------------------------------|------------|-------------------------------|-----------------------------------------------------|
| <b>HFrEF, HFmrEF, and HFpEF</b> |            |                               |                                                     |
| SGLT2 inhibitor                 | \$2172     | \$47 800 (\$17 200–\$79 800)  | \$6249                                              |
| ARNI                            | \$3589     | \$39 900 (\$21 800–\$60 300)  | \$10 975                                            |
| SGLT2 inhibitor + ARNI          | \$5761     | \$44 400 (\$35 000–\$54 300)  | \$15 510                                            |
| <b>HFrEF</b>                    |            |                               |                                                     |
| SGLT2 inhibitor                 | \$2172     | \$40 100 (\$21 500–\$59 400)  | \$8049                                              |
| ARNI                            | \$3589     | \$38 100 (\$23 200–\$53 800)  | \$11 670                                            |
| SGLT2 inhibitor + ARNI          | \$5761     | \$44 200 (\$34 400–\$54 200)  | \$15 510                                            |
| <b>HFmrEF and HFpEF</b>         |            |                               |                                                     |
| SGLT2 inhibitor                 | \$2172     | \$76 900 (\$30 900–\$121 700) | \$3631                                              |

ARNI indicates angiotensin receptor-neprilysin inhibitor; CMS, Centers for Medicare and Medicaid Services; EF, ejection fraction; HF, heart failure; HFmrEF, heart failure with mid-range ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; ICER, incremental cost-effectiveness ratio; QALY, quality-adjusted life year; SGLT2, sodium-glucose co-transporter 2; SOC, standard of care; and USD, US dollars.

\*Dapagliflozin price used for SGLT2 inhibitor.

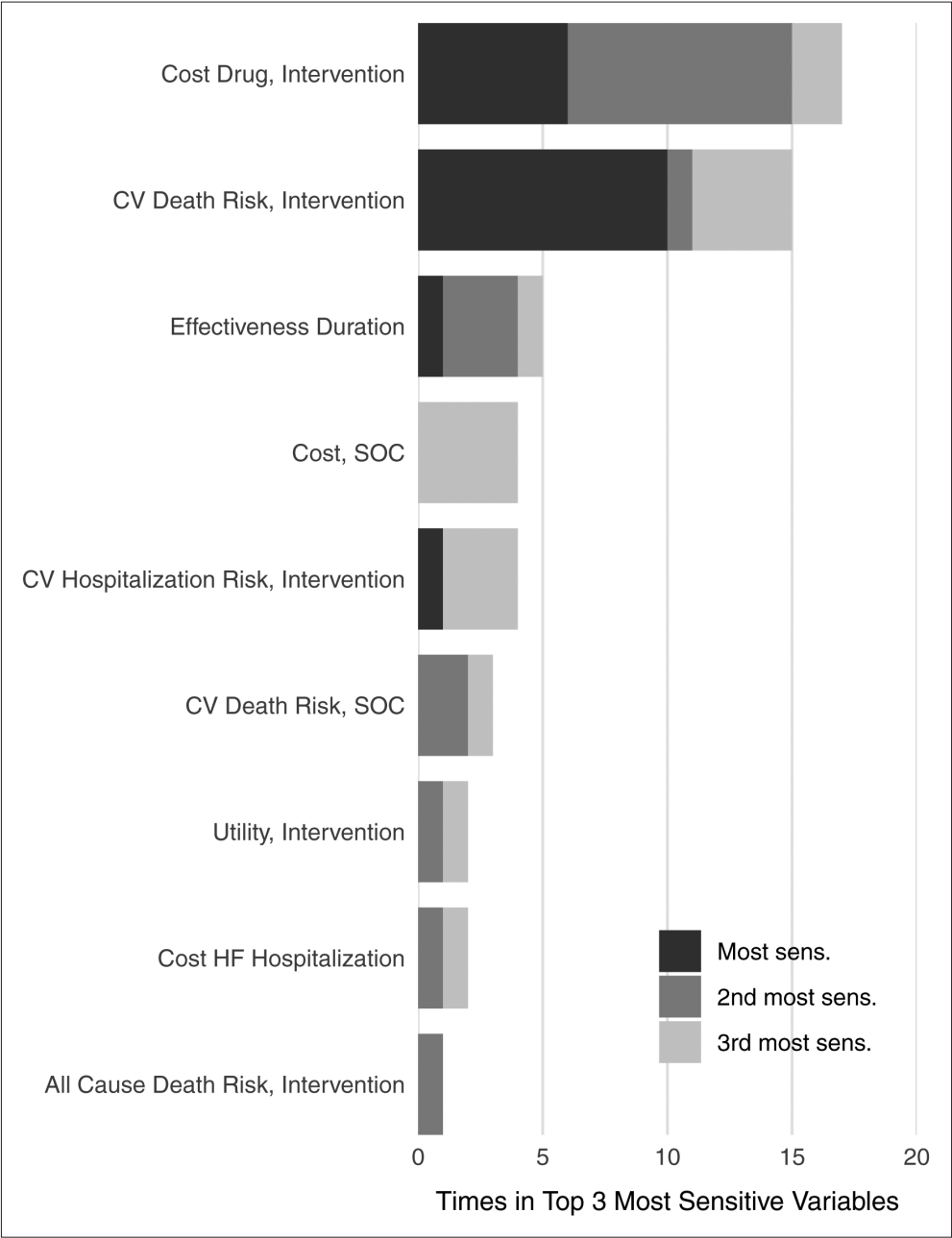
†The estimated ICER is derived from linear mixed-effects models with a random intercept for study and evaluated at the CMS-negotiated drug price. The reported 95% CIs reflect uncertainty in the estimated fixed-effects association between ICER and drug price across studies. The 95% CIs were derived using 500 parametric bootstrap samples from the fitted linear mixed-effects model. For each bootstrap replicate, fixed-effects predictions were generated (excluding random effects), and percentile-based 95% CIs were calculated from the 2.5th and 97.5th percentiles of the bootstrapped predictions.

‡The estimated drug price corresponding to the cost-effectiveness threshold of \$120 000/QALY gained was calculated by solving the fitted fixed-effects equation for drug price.

supports recent efforts to lower drug prices through mechanisms such as the Inflation Reduction Act and similar legislation that enables CMS to negotiate drug prices. These results bolster the case that drug price reductions are a critical lever to enhance value-based care, reduce patient cost-sharing, and improve access to high-value therapies. The CMS-negotiated prices would improve access to costly HF care while increasing economic efficiency. By expanding access, such reforms may also promote health equity, particularly for populations disproportionately affected by HF. Future cost-effectiveness analyses would benefit from considering the impacts on health equity of these costly drugs for HF, accounting for future CMS-negotiated prices. The observed variation in cost-effectiveness by LVEF suggests that expanding a therapy's indication may be an opportune moment to revisit pricing or reimbursement policies. Our analysis provides a framework to help decision-makers estimate cost-effectiveness as generic products become available and are more widely used and to inform future HF therapy price negotiations. It is important to assess cost-effectiveness over the life cycle of drug therapies, including expansion into generic products. As generic competition increases and prices decrease, our findings suggest that the cost-effectiveness of SGLT2 inhibitors and ARNIs will improve substantially. In addition, because one of the main drivers of cost-effectiveness for SGLT2 inhibitors

is their effect on cardiovascular mortality, and uncertainty remains in the HFmrEF population, future studies are needed to examine the long-term effects on mortality, specifically at higher LVEF ranges where cost-effectiveness is currently relatively lower.

This analysis can be used in future HF guidelines to inform the economic value statements, as recommended in the 2025 American Heart Association/American College of Cardiology Statement on Cost/Value Methodology in Clinical Practice Guidelines.<sup>32</sup> Economic value statements are recommended to contain (1) the class of recommendation (eg, cost-saving, cost-effective, not cost-effective), (2) the level of certainty (eg, high, moderate), and (3) a recommendation statement. For example, based on the data from this analysis, the following economic value statement could be made: Cost-effective (high level of certainty). In patients with HFrEF, the use of an SGLT2 inhibitor at a cost of \$4582 (2024 USD) is projected to be a cost-effective strategy compared with current SOC, with an ICER of \$73 500 per QALY gained. The intervention is projected to be cost-effective at a threshold of \$120 000 per QALY gained if the cost of the SGLT2 inhibitor is less than \$8049 (2024 USD). As recommended, Figure 3 could be included with the supporting text associated with this statement. More systematic reviews of the growing cost-effectiveness literature in cardiovascular conditions are needed to support clinical guidelines.



**Figure 4. Variables most frequently identified in 1-way sensitivity analyses.** The figure shows the number of times the categories of variables were in the top 3 most sensitive variables in 1-way sensitivity analyses. The darkest color indicates the number of times the category was identified as the most sensitive variable, middle color the second most sensitive, and lightest color the third most sensitive. CV indicates cardiovascular; HF, heart failure; sens., sensitive; and SOC, standard of care.

Strengths and Limitations

This study represents the most comprehensive review to date of the cost-effectiveness of SGLT2 inhibitors and ARNIs in the treatment of HF in the United States and considers recent changes in the coverage of these therapies. Although a systematic approach with multiple reviewers was used to reduce bias, choices to include or exclude reported secondary analyses in data collection and analysis could potentially introduce bias. A limitation of this study is that there is not agreement among

health economists as to the appropriateness and suitability of meta-analysis on cost-effectiveness analyses. Challenges with meta-analysis include lack of within-study variance estimates across outcomes and that an ICER is a ratio that approaches infinity when QALY differences approach zero. In this analysis, it was assumed that changes in drug prices would not affect QALY outcomes; therefore, the ICER would only reflect changes in the incremental costs because of the change in drug price. Due in part to a lack of variance estimates, we did not use formal meta-analytic approaches and rather

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summarized results using linear mixed-effects models, which accounted for between-study differences using a random intercept for study; this approach has its own limitations. Regardless, the association of drug prices with ICERs were observed to be linear in nearly all studies, and the approach provides an approximation of the estimated ICER with CMS-negotiated drug prices stratified by LVEF. CMS-negotiated prices currently apply only to Medicare beneficiaries, and extension of these pricing effects to commercial or Medicaid populations is assumed rather than observed. Furthermore, potential changes in adherence or uptake associated with lower drug prices were not modeled, which could alter both costs and QALYs in real-world practice. As one of the objectives was to examine the effect of CMS-negotiated prices, we chose to focus on cost-effectiveness studies of empagliflozin or dapagliflozin compared with SOC. However, there is additional evidence of the cost-effectiveness of SGLT2 inhibitors, including studies comparing sotagliflozin versus SOC<sup>36,37</sup> and head-to-head comparisons of SGLT2 inhibitors.<sup>34,35</sup> Both studies examining head-to-head comparisons of SGLT2 inhibitors found empagliflozin and dapagliflozin to be cost-effective (Table S3).<sup>34,35</sup> The studies examining sotagliflozin versus SOC were also found to be cost-effective, with ICERs of \$45 596 per QALY<sup>36</sup> and \$81 823 per QALY gained.<sup>37</sup> Estimates of cost-effectiveness should be updated as CMS price negotiations expand to include other medications. Another potential weakness is that this study did not weigh studies based on quality. The decision was made not to apply quality-weights to studies to avoid introducing bias based on the arbitrary nature of the decided weights.

## Conclusions

In this systematic review, the cost-effectiveness of SGLT2 inhibitors and ARNIs was examined for the treatment of HF across the spectrum of LVEF in the United States, in the context of recent CMS-negotiated prices resulting from the Inflation Reduction Act. In this regard, SGLT2 inhibitors and ARNIs demonstrated consistent cost-effectiveness in HF with reduced EF across pricing scenarios. For HF with mid-range EF and HF with preserved EF, cost-effectiveness estimates were fewer and more variable, with projected value dependent on drug price reductions under policies such as the Inflation Reduction Act and on underlying treatment-effect assumptions. These findings underscore both the established economic value of these therapies in HF with reduced EF and the evolving, model-dependent nature of economic projections in HF with mid-range EF and HF with preserved EF populations.

## ARTICLE INFORMATION

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## Supplemental Material

Final PubMed Search Strategy  
Protocol Deviations  
Tables S1–S4  
Figures S1–S3

## REFERENCES

- Bozkurt B, Ahmad T, Alexander KM, Baker WL, Bosak K, Brethett K, Fonarow GC, Heidenreich P, Ho JE, Hsieh E, et al; Writing Committee Members. Heart failure epidemiology and outcomes statistics: a report of the Heart Failure Society of America. *J Card Fail*. 2023;29:1412–1451. doi: 10.1016/j.cardfail.2023.07.006
- Heidenreich PA, Albert NM, Allen LA, Blumke DA, Butler J, Fonarow GC, Ikonomicis JS, Khavjou O, Konstam MA, Maddox TM, et al; American Heart Association Advocacy Coordinating Committee. Forecasting the impact of heart failure in the United States: a policy statement from the American Heart Association. *Circ Heart Fail*. 2013;6:606–619. doi: 10.1161/HHF.0b013e318291329a
- Wei C, Heidenreich PA, Sandhu AT. The economics of heart failure care. *Prog Cardiovasc Dis*. 2024;82:90–101. doi: 10.1016/j.pcad.2024.01.010
- McMurray JJV, Solomon SD, Inzucchi SE, Kober L, Kosiborod MN, Martinez FA, Ponikowski P, Sabatine MS, Anand IS, Belohlavek J, et al; DAPA-HF Trial Committees and Investigators. Dapagliflozin in patients with heart failure and reduced ejection fraction. *N Engl J Med*. 2019;381:1995–2008. doi: 10.1056/NEJMoa1911303
- Packer M, Anker SD, Butler J, Filippatos G, Pocock SJ, Carson P, Januzzi J, Verma S, Tsutsui H, Brueckmann M, et al; EMPEROR-Reduced Trial Investigators. Cardiovascular and renal outcomes with empagliflozin in heart failure. *N Engl J Med*. 2020;383:1413–1424. doi: 10.1056/NEJMoa2022190
- McMurray JJ, Packer M, Desai AS, Gong J, Lefkowitz MP, Rizkala AR, Rouleau JL, Shi VC, Solomon SD, Swedberg K, et al; PARADIGM-HF Investigators and Committees. Angiotensin-neprilysin inhibition versus enalapril in heart failure. *N Engl J Med*. 2014;371:993–1004. doi: 10.1056/NEJMoa1409077
- Solomon SD, de Boer RA, DeMets D, Hernandez AF, Inzucchi SE, Kosiborod MN, Lam CSP, Martinez F, Shah SJ, Lindholm D, et al. Dapagliflozin in heart failure with preserved and mildly reduced ejection fraction: rationale and design of the DELIVER trial. *Eur J Heart Fail*. 2021;23:1217–1225. doi: 10.1002/ehf.2249

8. Anker SD, Butler J, Filippatos G, Ferreira JP, Bocchi E, Bohm M, Brunner-La Rocca HP, Choi DJ, Chopra V, Chuquiere-Valenzuela E, et al; EMPEROR-Preserved Trial Investigators. Empagliflozin in heart failure with a preserved ejection fraction. *N Engl J Med*. 2021;385:1451–1461. doi: 10.1056/NEJMoa2107038
9. Vaduganathan M, Docherty KF, Claggett BL, Jhund PS, de Boer RA, Hernandez AF, Inzucchi SE, Kosiborod MN, Lam CSP, Martinez F, et al. SGLT-2 inhibitors in patients with heart failure: a comprehensive meta-analysis of five randomised controlled trials. *Lancet*. 2022;400:757–767. doi: 10.1016/S0140-6736(22)01429-5
10. Chandra A, Vaduganathan M, Lewis EF, Claggett BL, Rizkala AR, Wang W, Lefkowitz MP, Shi VC, Anand IS, Ge J, et al; PARAGON-HF Investigators. Health-related quality of life in heart failure with preserved ejection fraction: the PARAGON-HF trial. *JACC Heart Fail*. 2019;7:862–874. doi: 10.1016/j.jchf.2019.05.015
11. Yarmuth JHR. *Inflation reduction act of 2022*. 2025:5376. Accessed May 1, 2025. <https://www.congress.gov/bill/117th-congress/house-bill/5376>
12. Services CMM. *Medicare drug price negotiation program: negotiated prices for initial price applicability year 2026*. 2025. Accessed May 1, 2025. <https://www.congress.gov/bill/117th-congress/house-bill/5376>
13. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71. doi: 10.1136/bmj.n71
14. *Covidence systematic review software [computer program]*. www.covidence.org: Veritas Health Innovation. Accessed February 28, 2026. <https://support.covidence.org/help/how-can-i-cite-covidence>
15. King JB, Shah RU, Bress AP, Nelson RE, Bellows BK. Cost-effectiveness of sacubitril-valsartan combination therapy compared with enalapril for the treatment of heart failure with reduced ejection fraction. *JACC Heart Fail*. 2016;4:392–402. doi: 10.1016/j.jchf.2016.02.007
16. Sandhu AT, Ollendorf DA, Chapman RH, Pearson SD, Heidenreich PA. Cost-effectiveness of sacubitril-valsartan in patients with heart failure with reduced ejection fraction. *Ann Intern Med*. 2016;165:681–689. doi: 10.7326/M16-0057
17. Gaziano TA, Fonarow GC, Claggett B, Chan WW, Deschaseaux-Voinet C, Turner SJ, Rouleau JL, Zile MR, McMurray JJ, Solomon SD. Cost-effectiveness analysis of sacubitril/valsartan vs enalapril in patients with heart failure and reduced ejection fraction. *JAMA Cardiol*. 2016;1:666–672. doi: 10.1001/jamacardio.2016.1747
18. Zueger PM, Kumar VM, Harrington RL, Rigoni GC, Atwood A, DiDomenico RJ, Touchette DR. Cost-effectiveness analysis of sacubitril/valsartan for the treatment of heart failure with reduced ejection fraction in the United States. *Pharmacotherapy*. 2018;38:520–530. doi: 10.1002/phar.2108
19. Gaziano TA, Fonarow GC, Velazquez EJ, Morrow DA, Braunwald E, Solomon SD. Cost-effectiveness of sacubitril-valsartan in hospitalized patients who have heart failure with reduced ejection fraction. *JAMA Cardiol*. 2020;5:1236–1244. doi: 10.1001/jamacardio.2020.2822
20. Parizo JT, Goldhaber-Fiebert JD, Salomon JA, Khush KK, Spertus JA, Heidenreich PA, Sandhu AT. Cost-effectiveness of dapagliflozin for treatment of patients with heart failure with reduced ejection fraction. *JAMA Cardiol*. 2021;6:926–935. doi: 10.1001/jamacardio.2021.1437
21. Isaza N, Calvachi P, Raber I, Liu CL, Bellows BK, Hernandez I, Shen C, Gavin MC, Garan AR, Kazi DS. Cost-effectiveness of dapagliflozin for the treatment of heart failure with reduced ejection fraction. *JAMA Netw Open*. 2021;4:e2114501. doi: 10.1001/jamanetworkopen.2021.14501
22. Zheng J, Parizo JT, Spertus JA, Heidenreich PA, Sandhu AT. Cost-effectiveness of empagliflozin in patients with heart failure with preserved ejection fraction. *JAMA Intern Med*. 2022;182:1278–1288. doi: 10.1001/jamainternmed.2022.5010
23. Nguyen BN, Mital S, Bugden S, Nguyen HV. Cost-effectiveness of dapagliflozin and empagliflozin for treatment of heart failure with reduced ejection fraction. *Int J Cardiol*. 2023;376:83–89. doi: 10.1016/j.ijcard.2023.01.080
24. Bhatt AS, Vaduganathan M, Claggett BL, Fonarow GC, Packer M, Pfeffer MA, Shah SJ, Shen X, Cristino J, McMurray JJV, et al. Health and economic evaluation of sacubitril-valsartan for heart failure management. *JAMA Cardiol*. 2023;8:1041–1048. doi: 10.1001/jamacardio.2023.3216
25. Cohen LP, Isaza N, Hernandez I, Lewis GD, Ho JE, Fonarow GC, Kazi DS, Bellows BK. Cost-effectiveness of sodium-glucose cotransporter-2 inhibitors for the treatment of heart failure with preserved ejection fraction. *JAMA Cardiol*. 2023;8:419–428. doi: 10.1001/jamacardio.2023.0077
26. Yan BW, Spahillari A, Pandya A. Cost-effectiveness of quadruple therapy in management of heart failure with reduced ejection fraction in the United States. *Circ Cardiovasc Qual Outcomes*. 2023;16:e009793. doi: 10.1161/CIRCOUTCOMES.122.009793
27. Dixit NM, Parikh NU, Ziaieian B, Jackson N, Fonarow GC. Cost-effectiveness of comprehensive quadruple therapy for heart failure with reduced ejection fraction. *JACC Heart Fail*. 2023;11:541–551. doi: 10.1016/j.jchf.2023.01.004
28. Reifsnider OS, Tafazzoli A, Linden S, Ishak J, Rakonczai P, Stargardt M, Kuti E. Cost-effectiveness analysis of empagliflozin for treatment of patients with heart failure with reduced ejection fraction in the United States. *J Am Heart Assoc*. 2024;13:e029042. doi: 10.1161/JAHA.123.029042
29. Bhatt AS, Vaduganathan M, Claggett BL, Kulac IJ, Anand IS, Desai AS, Fang JC, Hernandez AF, Jhund PS, Kosiborod MN, et al. Cost effectiveness of dapagliflozin for heart failure across the spectrum of ejection fraction: an economic evaluation based on pooled, individual participant data from the DELIVER and DAPA-HF trials. *J Am Heart Assoc*. 2024;13:e032279. doi: 10.1161/JAHA.123.032279
30. Dixit NM, Truong KP, Vaduganathan M, Ziaieian B, Fonarow GC. Cost-effectiveness of medical therapy for heart failure with mildly reduced and preserved ejection fraction. *JACC Heart Fail*. 2024;12:1226–1237. doi: 10.1016/j.jchf.2024.03.006
31. Analysis USBE. *Personal consumption expenditures price index*. 2025. Accessed May 1, 2025. <https://www.bea.gov/data/personal-consumption-expenditures-price-index>
32. Kazi DS, Abdullah AR, Arnold SV, Basu A, Bellows BK, Brethett K, Chew DS, Cohen DJ, DeJong C, Heidenreich PA, et al; Writing Committee Members. 2025 AHA/ACC statement on cost/value methodology in clinical practice guidelines (update from 2014 statement): a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2025;152:e332–e358. doi: 10.1161/CIR.0000000000001377
33. Kim DD, Do LA, Synnott PG, Lavelle TA, Prosser LA, Wong JB, Neumann PJ. Developing criteria for health economic quality evaluation tool. *Value Health*. 2023;26:1225–1234. doi: 10.1016/j.jval.2023.04.004
34. Rane A, Nechi RN, Imam M, Zoni CR, Ndikumukiza C, Karaye IM, Yunusa I, Alanzi A. Comparative value of dapagliflozin vs empagliflozin in patients with heart failure and preserved ejection fraction: a cost-effectiveness analysis. *J Manag Care Spec Pharm*. 2023;29:1045–1053. doi: 10.18553/jmcp.2023.29.9.1045
35. Nechi RN, Rane A, Karaye RM, Ndikumukiza C, Alsahali S, Jatau AI, Zoni CR, Alanzi A, Karaye IM, Yunusa I. Cost-effectiveness of dapagliflozin vs empagliflozin for treating heart failure with reduced ejection fraction in the United States. *Clin Ther*. 2023;45:627–632. doi: 10.1016/j.clinthera.2023.05.002
36. Kim J, Wang S, Sikirica S, Shafrin J. Cost-effectiveness of sotagliflozin for the treatment of patients with diabetes and recent worsening heart failure. *J Comp Eff Res*. 2024;13:e230190. doi: 10.57264/ceer-2023-0190
37. Weintraub WS, Kolm P, Dolman S, Alva M, Bhatt DL, Zhang Z. Cost-effectiveness of sotagliflozin in SOLOIST-WHF. *JACC Heart Fail*. 2024;12:1600–1610. doi: 10.1016/j.jchf.2024.04.018
38. Nam K, Cho DS, Kim H, Kwon B, Yoon Y, Park C, Kim ES, Yoon JC, Park SK. Systematic review of the economic evaluation of sodium-glucose cotransporter-2 inhibitors used as treatment in patients with heart failure. *Clin Drug Investig*. 2023;43:463–474. doi: 10.1007/s40261-023-01283-6
39. Kuan WC, Sim R, Wong WJ, Dujaili J, Kasim S, Lee KK, Teoh SL. Economic evaluations of guideline-directed medical therapies for heart failure with reduced ejection fraction: a systematic review. *Value Health*. 2023;26:1558–1576. doi: 10.1016/j.jval.2023.05.011
40. Rezapour A, Azari S, Arabloo J, Kolivand P, Behzadifar M, Omid N, Sarabi Asiabar A, Saberian P, Pourasghari H, Bragazzi NL, et al. Cost-effectiveness of sacubitril/valsartan compared with enalapril in patients with heart failure with reduced ejection fraction: a systematic review. *J Tehran Heart Cent*. 2022;17:168–179. doi: 10.18502/jthc.v17i4.11603
41. Heidenreich PA, Bozkurt B, Aguilar D, Allen LA, Byun JJ, Colvin MM, Deswal A, Drazner MH, Dunlay SM, Evers LR, et al; ACC/AHA Joint Committee Members. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2022;145:e895–e1032. doi: 10.1161/CIR.0000000000001063